

TUTORIAL SOLUTIONS

[METAL1: DEFORMATION AND STRENGTHENING]

Updated on 2023-Dec-14

1. Determine the simplest form of:

(1) $\epsilon_{ijk}\delta_{kj}$

(2) $\epsilon_{3jk}a_ja_k$

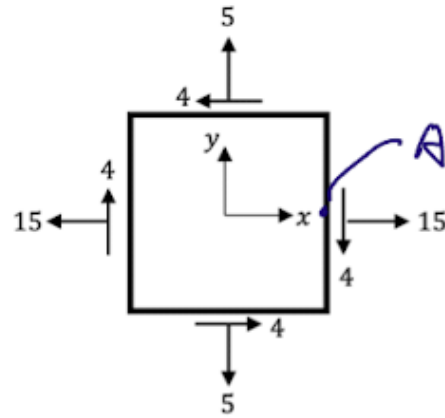
(3) $\epsilon_{1jk}\delta_{3j}b_k$

$$\epsilon_{ijk}\delta_{kj} = \epsilon_{ijj} = \epsilon_{i11} + \epsilon_{i22} + \epsilon_{i33} = 0$$

$$\begin{aligned}\epsilon_{3jk}a_ja_k &= \epsilon_{312}a_1a_2 + \epsilon_{321}a_2a_1 \\ &= \epsilon_{123}a_1a_2 - \epsilon_{123}a_1a_2 \\ &= 0\end{aligned}$$

$$\begin{aligned}\epsilon_{1jk}\delta_{3j}b_k &= \epsilon_{13k}b_k \\ &= \epsilon_{132}b_2 \\ &= -b_2\end{aligned}$$

2. Construct the Mohr's circles for the 2D stress state given by:



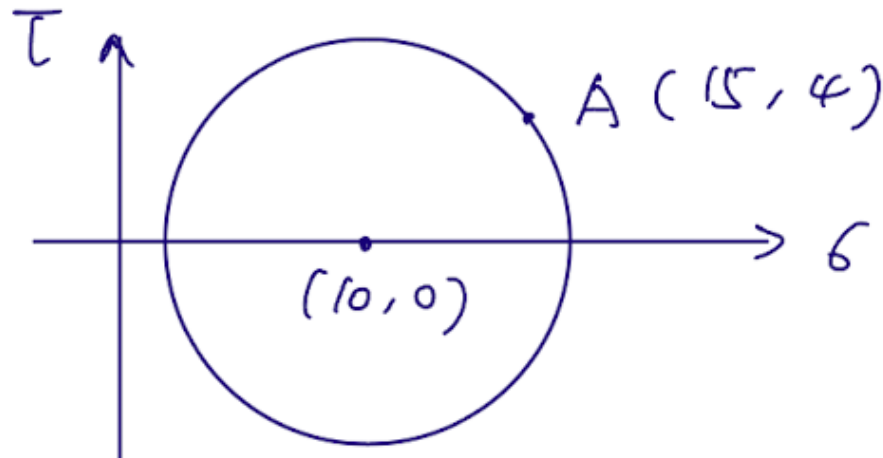
in units of MPa.

$$\sigma_{xx} = 15 \text{ MPa}$$

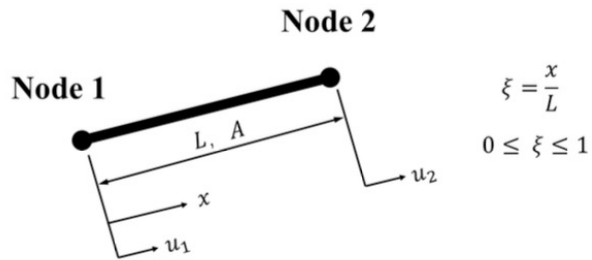
$$\sigma_{yy} = 5 \text{ MPa}$$

$$\sigma_{xy} = 4 \text{ MPa at point A}$$

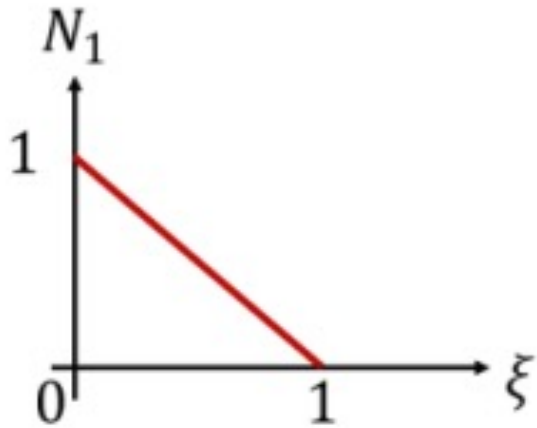
$$\begin{aligned}\sigma_{avg} &= \frac{\sigma_{xx} + \sigma_{yy}}{2} \\ &= \frac{15 \text{ MPa} + 5 \text{ MPa}}{2} \\ &= 10 \text{ MPa}\end{aligned}$$



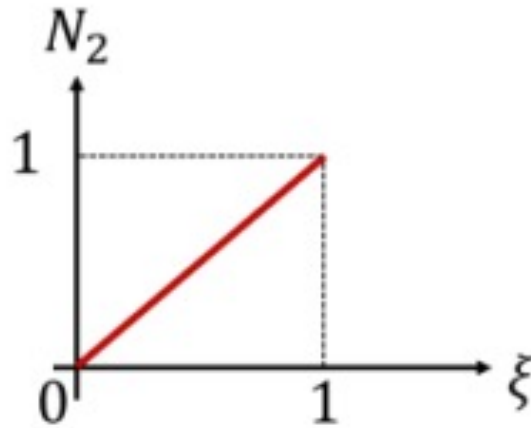
3. A two-noded linear truss element is shown.



(1) Write down shape functions N_1 and N_2 for nodes 1 and 2 in terms of the local spatial position ξ , and sketch the variation of each with ξ .



$$N_1(\xi) = 1 - \xi$$



$$N_2(\xi) = \xi$$

(2) The displacement along the element is given by

$$u = \mathbf{N} \cdot \mathbf{u}_I = (N_1 \quad N_2) \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

Determine the $\mathbf{B}$ matrix for this element in terms of the element length L .

$$\varepsilon = \mathbf{B} \cdot \mathbf{u}_I$$

$$\varepsilon = \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (\mathbf{N} \cdot \mathbf{u}_I) = \frac{\partial}{\partial x} (N_1 \quad N_2) \cdot \mathbf{u}_I$$

$$\varepsilon = \frac{\partial}{\partial \xi} (N_1 \quad N_2) \cdot \frac{\partial \xi}{\partial x} \cdot \mathbf{u}_I$$

$$\varepsilon = (-1 \quad 1) \cdot \frac{1}{L} \cdot \mathbf{u}_I$$

$$\therefore \mathbf{B} = \frac{1}{L} (-1 \quad 1)$$

- (3) Determine the stiffness matrix $\mathbf{k}$ for the element, given by $\mathbf{k} = \int_V \mathbf{B}^T \mathbf{D} \mathbf{B} dV$,
assuming the truss to be of cross-sectional area A and Young's modulus E .

$$\mathbf{k} = \int_V \mathbf{B}^T \mathbf{D} \mathbf{B} dV$$

$$\mathbf{k} = \int_L \mathbf{B}^T \mathbf{D} \mathbf{B} A dx$$

$$\mathbf{k} = \int_0^1 \mathbf{B}^T \mathbf{D} \mathbf{B} A L d\xi$$

$$\mathbf{k} = \int_0^1 \frac{1}{L} \begin{pmatrix} -1 \\ 1 \end{pmatrix} \cdot E \cdot \frac{1}{L} (-1 \ 1) \cdot A \cdot L d\xi$$

$$\therefore \mathbf{k} = \frac{EA}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

4. The following yield strengths were obtained in ferritic steel as a function of grain size.

Grain size (micron)	Yield strength (MPa)
40	180
12	280

- (1) Estimate the resistance of the lattice to dislocation motion, σ_0 , and the strengthening coefficient, k , in the Hall-Petch equation for this material and predict the expected yield strength of the steel in which the grain size is reduced to x micron.

$$\sigma_y = \sigma_0 + k \cdot d^{-0.5}$$

$$180\text{MPa} = \sigma_0 + k \cdot (40\mu\text{m})^{-0.5}$$

$$280\text{MPa} = \sigma_0 + k \cdot (12\mu\text{m})^{-0.5}$$

$$\sigma_0 = 58.90\text{MPa}$$

$$k = 765.92 \text{ MPa} \cdot (\mu\text{m})^{0.5}$$

$$\sigma_y = 58.90 + 765.92 \cdot x^{-0.5}$$

where x in micron and σ_y in MPa

(2) Name three methods to obtain smaller grain sizes in metal.

1. Treating (adding) the alloy with impurities called “Grain Refiners” (because they refine or reduce the grain size). These grain refiners increase the number of nucleation sites resulting in higher nucleation rates
2. Cooling the material faster during casting (liquid – solid transformation)
3. Cooling the material faster during heat treatment (solid – solid transformation)

(3) What are the three main heat treatment stages of precipitation hardening?

1. Solution heat treatment to form a homogeneous single phase solid solution.
2. Quenching to produce a supersaturated solid solution.
3. Aging to form fine dispersion of precipitates

5. Consider an aluminium alloy with FCC crystal structure. The shear modulus and the Poisson's ratio is given by $G = 27\text{GPa}$ and $\nu = 0.32$ respectively.

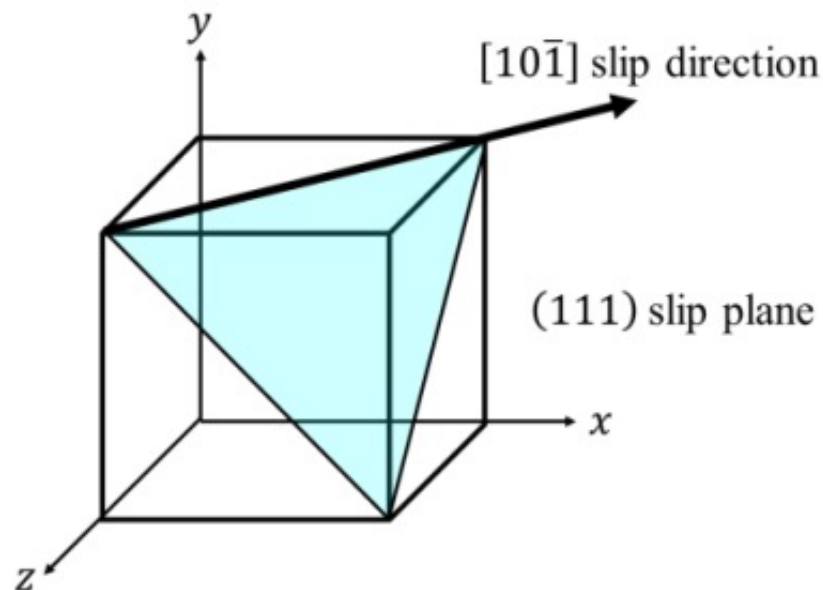
- (1) Write down the most commonly observed slip systems in the FCC structure (in the form of plane/direction family) and state how many different slip systems there are in that family.

The most commonly observed slip systems in FCC structure are

$$\{111\} \langle 110 \rangle$$

There are 12 different slip systems in that family

- (2) Sketch one slip system of that family with the coordinate axes and slip plane/direction clearly labeled.



- (3) Determine the total strain energy per unit length of a straight edge and a straight screw dislocation if the dislocation core radius is 0.8nm, the screening length is 8μm and the magnitude of Burgers vector $b = 0.25\text{nm}$.

Note:

The total strain energy of a strain dislocation is given by

$$E_{total} = \frac{Gb^2L}{4\pi K} \left[\ln \left(\frac{R}{r_0} \right) + 2K \right]$$

$$1\text{eV} = 1.6 \times 10^{-19}\text{J}$$

$$1\text{J} = 1\text{Pa} \cdot \text{m}^3$$

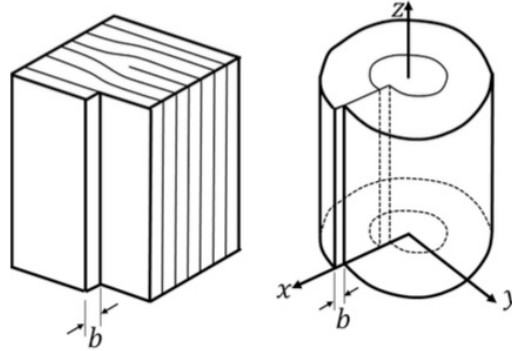
For a straight screw dislocation:

$$\begin{aligned} K &= 1, \quad R = 8\mu\text{m}, \quad r_0 = 0.8\text{nm} \\ \frac{E_{total}}{L} &= \frac{27\text{GPa} \times (0.25\text{nm})^2}{4\pi} \left[\ln \left(\frac{8\mu\text{m}}{0.8\text{nm}} \right) + 2 \right] \\ \frac{E_{total}}{L} &= 1.5054 \times 10^{-9}\text{J/m} = 9.4 \times 10^9\text{eV/m} \end{aligned}$$

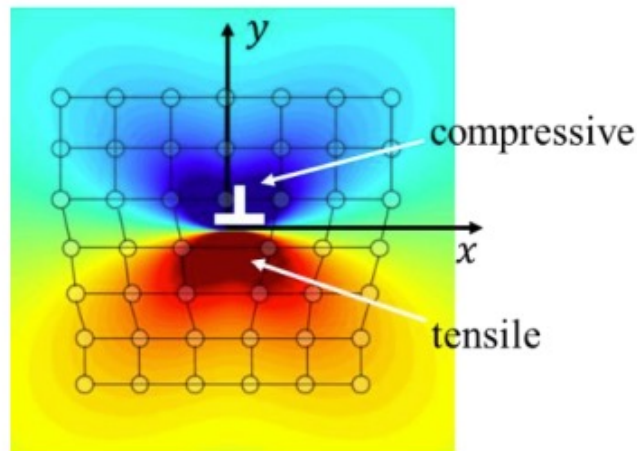
For a straight edge dislocation:

$$\begin{aligned} K &= 1 - \nu, \quad R = 8\mu\text{m}, \quad r_0 = 0.8\text{nm} \\ \frac{E_{total}}{L} &= \frac{27\text{GPa} \times (0.25\text{nm})^2}{4\pi \times (1 - 0.32)} \left[\ln \left(\frac{8\mu\text{m}}{0.8\text{nm}} \right) + 2 \times (1 - 0.32) \right] \\ \frac{E_{total}}{L} &= 2.087 \times 10^{-9}\text{J/m} = 1.3 \times 10^{10}\text{eV/m} \end{aligned}$$

6. The elastic distortion around an infinitely-long, straight edge dislocation can be represented in terms of a cylinder of elastic material shown below. The shear modulus and the Poisson's ratio is given by G and ν respectively. The magnitude of the Burgers vector is b .



- (1) Sketch the stress distribution of σ_{xx} in the $x - y$ plane. Label the compressive and tensile stress region.



Correct sketch for [3'], correct label of compressive and tensile stresses for [2']

(2) Calculate the effective pressure on a volume element of that cylinder using the non-zero stress components.

Note: the non-zero components of the stress field out of the dislocation core at location (x, y) are given by

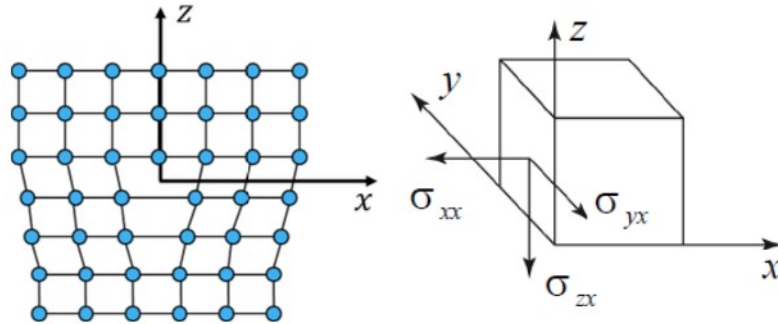
$$\begin{aligned}\sigma_{xx} &= -\frac{Gb}{2\pi(1-\nu)} y \frac{(3x^2 + y^2)}{(x^2 + y^2)^2} \\ \sigma_{yy} &= \frac{Gb}{2\pi(1-\nu)} y \frac{(x^2 - y^2)}{(x^2 + y^2)^2} \\ \sigma_{xy} = \sigma_{yx} &= \frac{Gb}{2\pi(1-\nu)} x \frac{(x^2 - y^2)}{(x^2 + y^2)^2} \\ \sigma_{zz} &= \nu(\sigma_{xx} + \sigma_{yy})\end{aligned}$$

$$p = -\frac{1}{3}(\sigma_{xx} + \sigma_{yy} + \sigma_{zz})$$

$$p = -\frac{1}{3} \times \frac{Gb}{2\pi(1-\nu)} [\sigma_{xx} + \sigma_{yy} + \nu(\sigma_{xx} + \sigma_{yy})]$$

$$p = \frac{Gb(1+\nu)}{3\pi(1-\nu)} \frac{y}{(x^2 + y^2)}$$

- (3) Assume the Burgers vector of an edge dislocation showing below is given by $\mathbf{b} = (-b, 0, 0)$ and the line direction is defined as $\mathbf{l} = (0, 1, 0)$. The Peach-Koehler force is then given by $\mathbf{F} = (\boldsymbol{\sigma} \cdot \mathbf{b}) \times \mathbf{l}$ under stress $\boldsymbol{\sigma}$. Determine the magnitude of each component of the Peach-Koehler force and their contributions to the movement of the considered dislocation. You may define a vector $\mathbf{g} = \boldsymbol{\sigma} \cdot \mathbf{b}$ to help with the calculation of $\mathbf{F}$.



glide plane is $(0, 0, 1)$, Burgers vector is $\mathbf{b} = (-b, 0, 0)$, line direction is $\mathbf{l} = (0, 1, 0)$

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}$$

$$g_x = \sigma_{xx}b_x + \sigma_{xy}b_y + \sigma_{xz}b_z = -\sigma_{xx}b$$

$$g_y = \sigma_{yx}b_x + \sigma_{yy}b_y + \sigma_{yz}b_z = -\sigma_{yx}b$$

$$g_z = \sigma_{zx}b_x + \sigma_{zy}b_y + \sigma_{zz}b_z = -\sigma_{zx}b$$

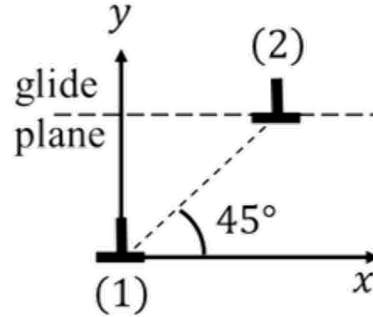
Using the right-hand convention of cross product:

$$g_x = -\sigma_{xx}b \text{ perpendicular to } \mathbf{l} \rightarrow \text{climb force, } |F_z| = \sigma_{xx}b$$

$$g_y = -\sigma_{yx}b \text{ parallel to } \mathbf{l} \rightarrow \text{makes no contribution to } \mathbf{F}, |F_y| = 0$$

$$g_z = -\sigma_{zx}b \text{ perpendicular to } \mathbf{l} \rightarrow \text{glide force, } |F_x| = \sigma_{zx}b$$

- (4) Consider two edge dislocations shown below. (Both axes have the same scale.) Dislocation (1) is pinned by a strong obstacle at (0,0). Is dislocation (2) at equilibrium position at room temperature? If so, is it stable? Please justify your answer.



Note: For dislocation (2), the force generated by dislocation (1) is given by

$$\mathbf{F} = \begin{pmatrix} \frac{Gb_1b_2}{2\pi(1-\nu)} \frac{x(x^2 - y^2)}{(x^2 + y^2)^2} \\ \frac{Gb_1b_2}{2\pi(1-\nu)} \frac{y(3x^2 + y^2)}{(x^2 + y^2)^2} \\ 0 \end{pmatrix}^T$$

Dislocation (2) is at equilibrium position at room temperature

Dislocation (2) is at unstable equilibrium position.

For dislocation (2), there is $x = y$, hence the force generated by dislocation (1) is given by

$$\mathbf{F} = \begin{pmatrix} 0 \\ \frac{Gb_1b_2}{2\pi(1-\nu)} \frac{1}{y} \\ 0 \end{pmatrix}^T$$

If $0 < x < y$, $F_x < 0$, attraction;

If $x > y$, $F_x > 0$, repulsion;

Any fluctuation of the stress will remove dislocation (2) from its position, hence it is unstable.

7.

(1) State how to calculate engineering strain and true strain.

The engineering strain or nominal strain $e = \frac{l-l_0}{l_0} = \frac{\Delta l}{l_0}$

The true strain or logarithmic strain $\varepsilon = \ln \left(\frac{l}{l_0} \right) = \ln (1 + e)$

(2) A bar 10cm long is elongated to 20cm by rolling in three steps: 10cm to 12cm; 12cm to 15cm, and 15 to 20cm. Calculate the engineering strain and true strain for each step and compare the sum of these with the overall deformation in each case.

By engineering strain

$$e_1 = \frac{2}{10} = 0.2; e_2 = \frac{3}{12} = 0.25; e_3 = \frac{5}{15} = 0.333;$$

Sum of e_1, e_2 and e_3

$$e_{sum} = 0.2 + 0.25 + 0.333 = 0.833$$

overall engineering strain:

$$e_{all} = \frac{10}{10} = 1$$

By true strain

$$\varepsilon_1 = \ln\left(\frac{12}{10}\right) = 0.182; \varepsilon_2 = \ln\left(\frac{15}{12}\right) = 0.223; \varepsilon_3 = \ln\left(\frac{20}{15}\right) = 0.288$$

Sum of $\varepsilon_1, \varepsilon_2$ and ε_3

$$\varepsilon_{sum} = 0.182 + 0.223 + 0.288 = 0.693$$

overall true strain:

$$\varepsilon_{all} = \ln \frac{20}{10} = 0.693$$

(3) Are true strains or engineering strains additive?

The true strain.

8. A state of stress in tensor notation is expressed as

$$\sigma = \begin{bmatrix} 0 & 172 & 0 \\ 172 & 0 & 0 \\ 0 & 0 & 100 \end{bmatrix} (\text{MPa})$$

(1) Find the principal stresses σ_1 , σ_2 and σ_3 .

$$I_1 = \sigma_{xx} + \sigma_{yy} + \sigma_{zz} = 100$$

$$I_2 = \sigma_{yz}^2 + \sigma_{zx}^2 + \sigma_{xy}^2 - \sigma_{yy}\sigma_{zz} - \sigma_{zz}\sigma_{xx} - \sigma_{xx}\sigma_{yy} = 172^2$$

$$I_3 = \sigma_{xx}\sigma_{yy}\sigma_{zz} + 2\sigma_{yz}\sigma_{zx}\sigma_{xy} - \sigma_{xx}\sigma_{yz}^2 - \sigma_{yy}\sigma_{zx}^2 - \sigma_{zz}\sigma_{xy}^2 = -100 \times 172^2$$

The magnitudes of the principal stresses σ_p , are the roots of

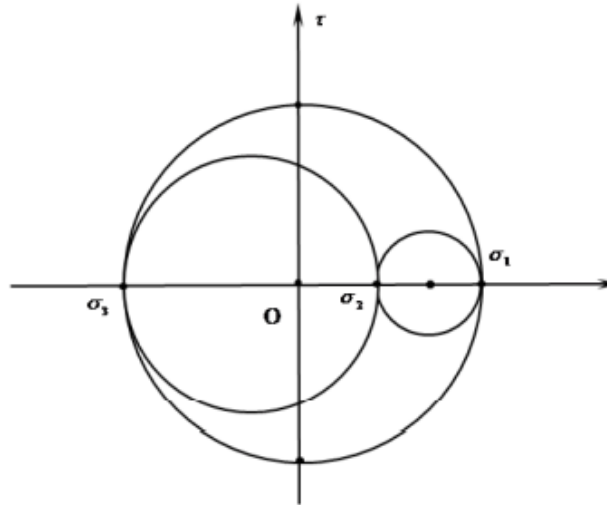
$$\sigma_p^3 - I_1\sigma_p^2 - I_2\sigma_p - I_3 = 0$$

the principal stresses are the roots,

$$\sigma_1 = 172(\text{MPa}), \sigma_2 = 100(\text{MPa}) \text{ and } \sigma_3 = -172(\text{MPa})$$

(2) Sketch the Mohr's circles for the stress state above.

The intersection points are: $(-172,0)$, $(100,0)$ and $(172,0)$, the radius are 172, 136 and 36



(3) If the material has a yield strength of 330MPa, use the von Mises criteria to determine whether the state of stress causes yielding.

Von mises criteria:

$$\bar{\sigma} = \frac{1}{\sqrt{2}} \sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2} = 314.2 < 330$$

Thus, the state of stress doesn't cause yielding.

9. Consider a tensor $\mathbf{S}$ and a vector $\mathbf{w}$ related as follows:

$$S_{kj} = \epsilon_{ijk} w_i$$

where ϵ_{ijk} is the permutation tensor.

(1) Write out $\mathbf{S}$ as a 3×3 matrix.

$$\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = 1 \quad \epsilon_{321} = \epsilon_{213} = \epsilon_{132} = -1$$

$$S_{11} = S_{22} = S_{33} = 0$$

$$S_{12} = \epsilon_{121} w_1 + \epsilon_{221} w_2 + \epsilon_{321} w_3 = -w_3$$

$$S_{13} = \epsilon_{131} w_1 + \epsilon_{231} w_2 + \epsilon_{331} w_3 = w_2$$

$$S_{21} = \epsilon_{112} w_1 + \epsilon_{212} w_2 + \epsilon_{312} w_3 = w_3$$

$$S_{23} = \epsilon_{132} w_1 + \epsilon_{232} w_2 + \epsilon_{332} w_3 = -w_1$$

$$S_{31} = \epsilon_{113} w_1 + \epsilon_{213} w_2 + \epsilon_{313} w_3 = -w_2$$

$$S_{32} = \epsilon_{123} w_1 + \epsilon_{223} w_2 + \epsilon_{323} w_3 = w_1$$

$$\underline{\underline{\mathbf{S}}} = \begin{bmatrix} 0 & -w_3 & w_2 \\ w_3 & 0 & -w_1 \\ -w_2 & w_1 & 0 \end{bmatrix}$$

(2) Show, using index notation, that $\underline{S}$ is a skew symmetric tensor ($\underline{S} = -\underline{S}^T$).

for the permutation tensor $\epsilon_{ijk} = -\epsilon_{ikj}$

$$S_{kj} = \epsilon_{ijk} w_i = -\epsilon_{ikj} w_i = -S_{jk}$$

$$\text{Hence, } \underline{\underline{S}} = -\underline{\underline{S}}^T$$

(3) Show that for any vector, $\underline{u}$,

$$\underline{S} \cdot \underline{u} = \underline{w} \times \underline{u}$$

for the permutation tensor $\epsilon_{ijk} = \epsilon_{kij}$

$$(\underline{\underline{S}} \cdot \underline{\underline{u}})_k = S_{kj} u_j = \epsilon_{ijk} w_i u_j$$

$$(\underline{w} \times \underline{u})_k = \epsilon_{kij} w_i u_j$$

Same

$$\text{Hence } (\underline{\underline{S}} \cdot \underline{\underline{u}})_k = (\underline{w} \times \underline{u})_k, \quad \underline{\underline{S}} \cdot \underline{\underline{u}} = \underline{w} \times \underline{u}$$

10. Consider a uniform bar made from a linear elastic material with Young's modulus E , length L and cross sectional area A . The bar is fixed at $x = 0$ and is subjected to a point load of magnitude F at $x = L$.

- (1) Assuming that this problem is to be modelled using a two-noded linear finite element. Find the $\mathbf{B}$ matrix for the element using the shape functions.

shape functions $N_1(\xi) = 1 - \xi$, $N_2(\xi) = \xi$, $\xi = \frac{x}{L}$

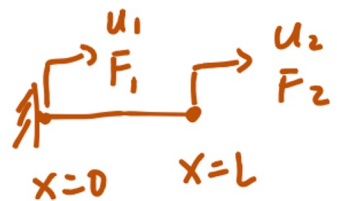
$$\underline{\mathbf{B}} = \frac{\partial}{\partial x} (N_1 \ N_2) = \frac{\partial}{\partial \xi} (N_1 \ N_2) \frac{\partial \xi}{\partial x}$$

$$= (-1 \quad 1) \cdot \frac{1}{L} \quad \text{Hence, } \underline{\mathbf{B}} = \frac{1}{L} (-1 \quad 1)$$

(2) Determine the stiffness matrix $\mathbf{k}$ for the element, given by $\mathbf{k} = \int_V \mathbf{B}^T \mathbf{D} \mathbf{B} dV$.

$$\begin{aligned}\underline{\mathbf{K}} &= \int_V \underline{\mathbf{B}}^T \mathbf{D} \underline{\mathbf{B}} dV = \int_L \underline{\mathbf{B}}^T \mathbf{D} \underline{\mathbf{B}} \cdot A dx = \int_0^1 \underline{\mathbf{B}}^T \mathbf{D} \underline{\mathbf{B}} \cdot AL d\xi \\ &= \int_0^1 \frac{1}{L} \begin{pmatrix} -1 \\ 1 \end{pmatrix} \cdot E \cdot \frac{1}{L} \begin{pmatrix} -1 & 1 \end{pmatrix} \cdot A \cdot L \cdot d\xi \\ &= \frac{EA}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\end{aligned}$$

(3) Compute the nodal force vector, apply appropriate boundary conditions and solve for the unknown displacements at the nodes.


$$\begin{array}{lll} u_1 = 0 & u_2 = u \\ F_1 = ? & F_2 = F & \underline{\mathbf{f}}_i = \underline{\mathbf{K}} \cdot \underline{\mathbf{u}}_i \end{array}$$

$$\begin{pmatrix} F_1 \\ F \end{pmatrix} = \frac{EA}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ u \end{pmatrix}$$

$$\bar{F}_1 = -F \quad u = \frac{FL}{EA}$$

11.

(1) The stress tensor at a point P is given by:

$$\sigma = \begin{pmatrix} 700 & 100 & -20 \\ 100 & 500 & 0 \\ -20 & 0 & 410 \end{pmatrix}$$

in units of MPa. Determine the traction at P on the plane with unit normal given by:

$$n = \frac{1}{3}(2, -1, 2)$$

$$\underline{t} = \underline{\sigma} \cdot \underline{n} = \begin{pmatrix} 700 & 100 & -20 \\ 100 & 500 & 0 \\ -20 & 0 & 410 \end{pmatrix} \cdot \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$
$$= \begin{pmatrix} 420 \\ -100 \\ 260 \end{pmatrix}$$

(2) Evaluate the invariants I_1 , I_2 and I_3 of the stress tensor:

$$\sigma = s \begin{pmatrix} 0.75 & -0.375 & 0 \\ -0.375 & 0.75 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

where s is a constant in units of MPa. Verify that one of the principal stresses is s , and find the other two in terms of s .

$$I_1 = \text{tr}(\underline{\underline{\sigma}}) = 2.5s \quad I_2 = -1.922s^2 \quad I_3 = \det(\underline{\underline{\sigma}}) = 0.422s^3$$

$$\sigma^3 - I_1 \sigma^2 - I_2 \sigma - I_3 = 0$$

$$\sigma^3 - 2.5s \sigma^2 + 1.922s^2 \sigma - 0.422s^3 = 0$$

$$(\sigma - s)(\sigma^2 - 1.5s\sigma + 0.422s^2) = 0$$

$$\sigma = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sigma_1 = 1.125s$$

$$\sigma_2 = s$$

$$\sigma_3 = 0.375s$$

(3) Construct the Mohr's circles for the stress state given by:

$$\sigma = \begin{pmatrix} 5 & 0 & -5 \\ 0 & -5 & 0 \\ -5 & 0 & 5 \end{pmatrix}$$

in units of MPa.

$$I_1 = 5 \quad I_2 = 50 \quad I_3 = 0$$

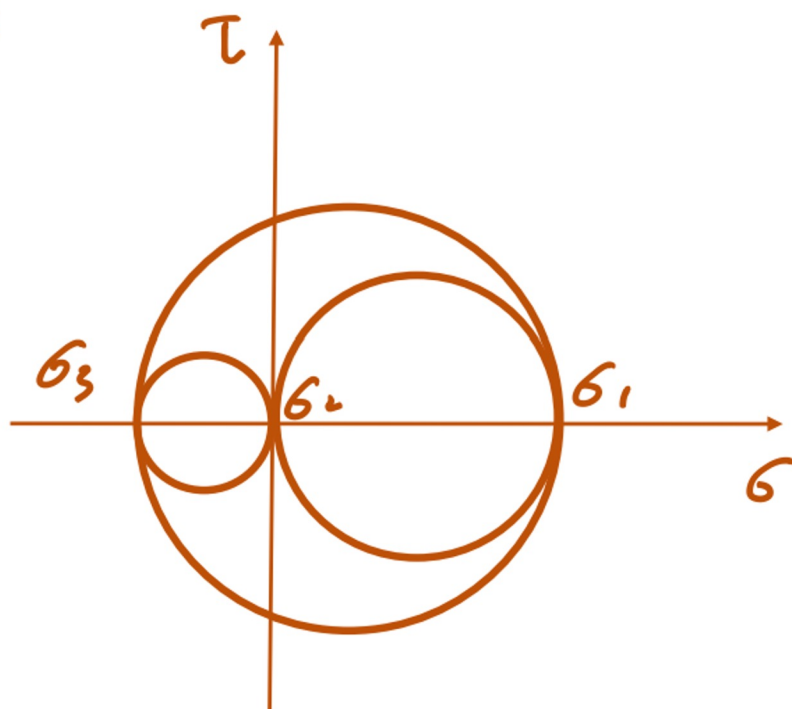
$$\sigma^3 - I_1 \sigma^2 - I_2 \sigma - I_3 = 0$$

$$\sigma^3 - 5\sigma^2 - 50\sigma = 0$$

$$\sigma_1 = 10$$

$$\sigma_2 = 0$$

$$\sigma_3 = -5$$



(4) Split the stress tensor given by:

$$\sigma = \begin{pmatrix} 120 & 40 & 0 \\ 40 & 90 & -20 \\ 0 & -20 & 30 \end{pmatrix}$$

into its hydrostatic and deviatoric parts. Show that the first invariant of the deviatoric part is zero.

$$\sigma_h = \frac{\sigma_{11} + \sigma_{22} + \sigma_{33}}{3} = 80$$

$$\underline{\underline{\sigma_s}} = \underline{\underline{\sigma}} - \sigma_h \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 40 & 40 & 0 \\ 40 & 10 & -20 \\ 0 & -20 & -50 \end{pmatrix}$$

$$I_1(\underline{\underline{\sigma_s}}) = \text{tr}(\underline{\underline{\sigma_s}}) = 40 + 10 - 50 = 0$$

12. Consider a bar with the original length of 10cm and is elongated to 20cm by rolling in three steps: 10cm to 12cm; 12cm to 15cm, and 15 to 20cm.
- (1) Calculate the engineering strain and true strain for each step and compare the sum of these with the overall deformation.
 - (2) Are true strains or engineering strains additive?
 - (3) What are the two prominent intragranular mechanisms of cold plastic deformation?

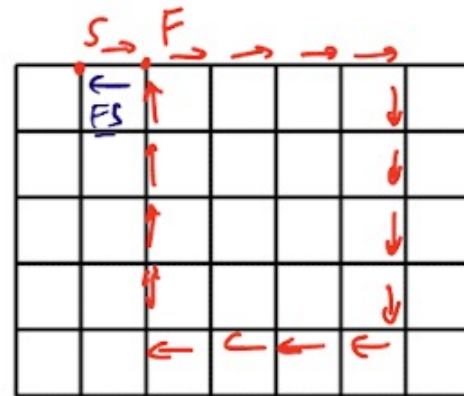
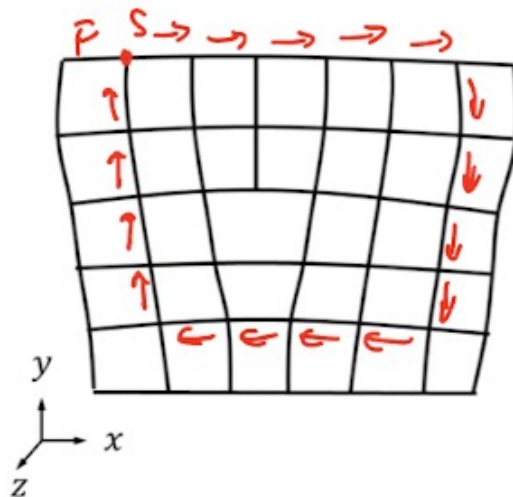
slip and twinning

- (4) State the key differences between the Tresca and Von Mises yield criterion.

The Tresca yield criterion does not consider the effect of the intermediate stress. The order of the three principal stresses is unknown. The Von Mises criterion has considered the intermediate stresses.

13. Consider a pure nickel specimen with the FCC crystal structure.

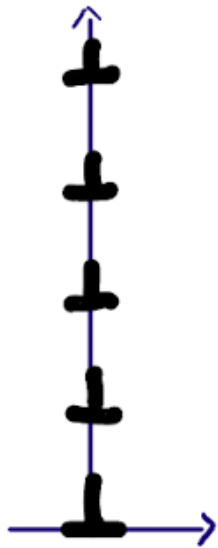
- (1) What are the most commonly observed slip systems in this material (in the form of plane/direction family) and how many different slip systems are in this family?
- (2) Sketch one slip plane and all the potential slip directions on this plane of the FCC structure. Please label the coordinate axes and the corresponding slip plane and directions.
- (3) Consider a straight edge dislocation that exists in the specimen. Use the following two figures and define the positive direction of the dislocation line to find the expression of the Burgers vector according to the RHFS convention. Note the magnitude of the Burgers vector can be denoted by b .



Define $\underline{l} = (0, 0, -1)$ $\underline{FS} = \underline{b} = (-b, 0, 0)$

- (4) Give an example of the stable equilibrium arrangement of the same-signed parallel straight edge dislocations. Please sketch this dislocation arrangement.
- (5) Does the configuration in the previous question have a long-range stress field? Is the average energy of the dislocation in this dislocation array higher or lower than that of an individual dislocation?

dislocation wall



Yes, it has a long-range stress field.

The energy is lower than the energy of an individual dislocation

14. Prove the following identities **using index notation**:

(1) For vectors, $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$,

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$$

$$(\underline{u} \times \underline{v}) \cdot \underline{w} = (\underline{u} \times \underline{v})_i w_i = \epsilon_{ijk} u_j v_k w_i$$

$$\underline{u} \cdot (\underline{v} \times \underline{w}) = u_i (\underline{v} \times \underline{w})_i = \epsilon_{ijk} u_i v_j w_k$$

$$= \epsilon_{kij} u_i v_j w_k \text{ (let } i=j, j=k, k=i)$$

$$= \epsilon_{ijk} u_j v_k w_i = (\underline{u} \times \underline{v}) \cdot \underline{w}$$

(2) For two second order tensors $\mathbf{A}$ and $\mathbf{B}$,

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$$

$$\text{tr}(C_{ik}) = C_{ii}$$

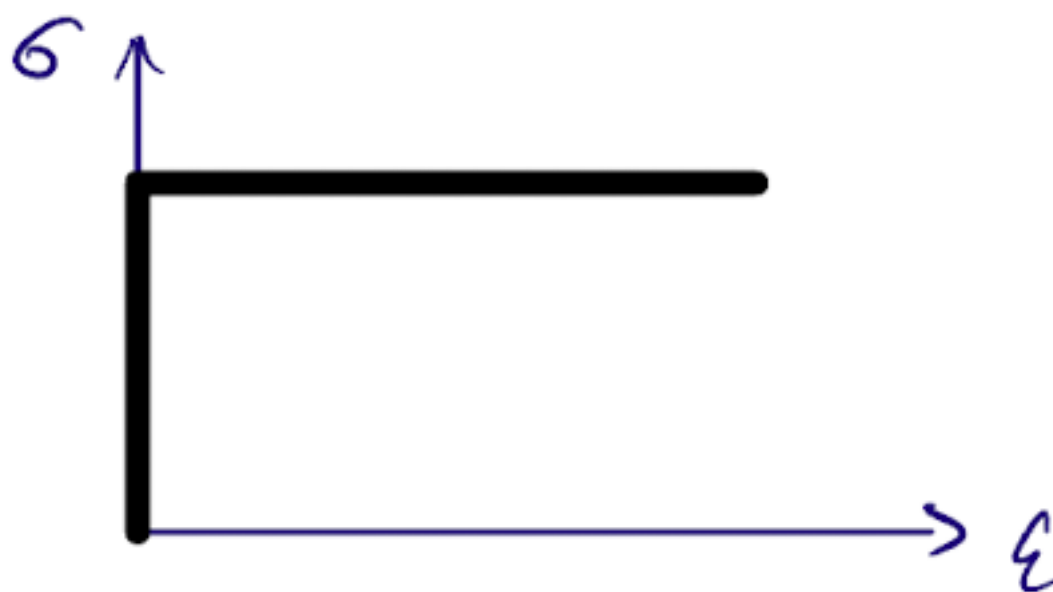
$$\text{tr}(\underline{\underline{A}} \underline{\underline{B}}) = \text{tr}(A_{ij} B_{jk}) = A_{ij} B_{ji}$$

$$\text{tr}(\underline{\underline{B}} \underline{\underline{A}}) = \text{tr}(B_{ij} A_{jk}) = B_{ij} A_{ji}$$

$$= A_{ji} B_{ij} \quad (\text{let } i=j, j=i)$$

$$= A_{ij} B_{ji} = \text{tr}(\underline{\underline{A}} \underline{\underline{B}})$$

15. Sketch a typical stress-strain curve for perfect rigid-plastic materials.



16. Calculate the hydrostatic and deviatoric stress for the biaxial tension shown below.↵

$$\sigma = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 8 \end{bmatrix} \text{↵}$$

in units of MPa.↵

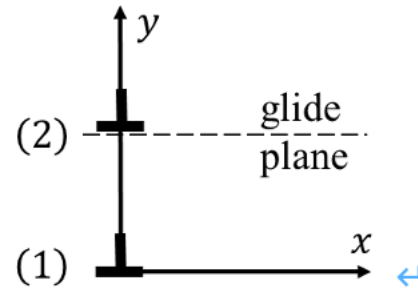
$$\sigma_m = \frac{\sigma_{ii}}{3} = \frac{10+8}{3} = 6 \text{ MPa}$$

$$\underline{\underline{S}} = \underline{\underline{\sigma}} - \sigma_m \underline{\underline{I}}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 8 \end{bmatrix} - 6 \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -6 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix} \text{ MPa}$$

17. Consider two edge dislocations shown below. Dislocation (1) is pinned by a strong obstacle at (0,0). Is dislocation (2) at an equilibrium position at room temperature? If so, is it stable? Please justify your answer. ←



Note: For dislocation (2), the force generated by dislocation (1) is given by ←

$$\mathbf{F} = \begin{pmatrix} \frac{Gb_1b_2}{2\pi(1-\nu)} \frac{x(x^2+y^2)}{(x^2+y^2)^2} \\ \frac{Gb_1b_2}{2\pi(1-\nu)} \frac{y(3x^2+y^2)}{(x^2+y^2)^2} \\ 0 \end{pmatrix}^T \quad \leftarrow$$

Disl. (2) is at equilibrium

Disl. (2) is stable

for disl. (2), $x=0$, hence the force on (2) due to disl. (1) is given by

$$\underline{F} = \left(0, \frac{Gb_1b_2}{2\pi(1-\nu)y}, 0 \right)$$

if $-y < x < 0$, $F_x > 0$, attraction

if $0 < x < y$, $F_x < 0$, attraction

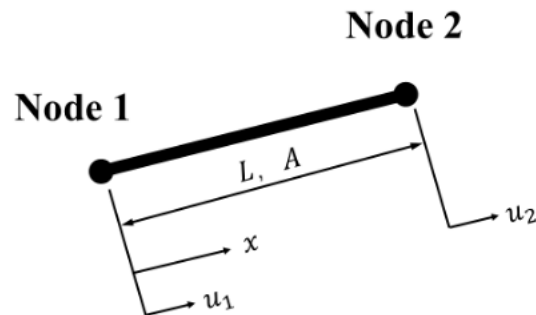
Any fluctuation will move disl. (2) back to its position,

hence it is stable

18. A uniaxial tension bar of length 300mm and radius 5mm made of mild steel ($E=210\text{GPa}$) carries a load of 3kN. Using a single truss finite element shown below to model the bar, determine the finite element stiffness matrix given by ↵

$$\underline{k} = \frac{EA}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \leftarrow$$

By specifying the nodal force vector and using the stiffness matrix, determine the extension of the tension bar. ↵



$$\xi = \frac{x}{L}$$

$$0 \leq \xi \leq 1$$

$$\underline{k} = \frac{EA}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$= \frac{210 \times 10^9 \text{ Pa} \times \pi \times (5 \times 10^{-3})^2 \text{ m}^2}{300 \times 10^{-3} \text{ m}} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$= 55 \times 10^6 \times \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \text{ Pa} \cdot \text{m} = \text{N} \cdot \text{m}^{-1}$$

$$\underline{f} = \underline{k} \cdot \underline{u}$$

$$\begin{pmatrix} f_1 \\ f_2 \end{pmatrix} = \underline{k} \cdot \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \quad \begin{matrix} f_2 = 3 \text{ kN} \\ u_1 = 0 \end{matrix}$$

$$3 \times 10^3 \text{ N} = 55 \times 10^6 \times u_2 \text{ (N} \cdot \text{m}^{-1})$$

$$f_1 = -55 \times 10^6 \times (-u_2) \text{ (N} \cdot \text{m}^{-1})$$

$$f_1 = -3 \text{ kN}$$

$$u_2 = 0.0545 \times 10^{-3} \text{ m} = 0.0545 \text{ mm} = 54.5 \mu\text{m}$$